

Using Intercepts

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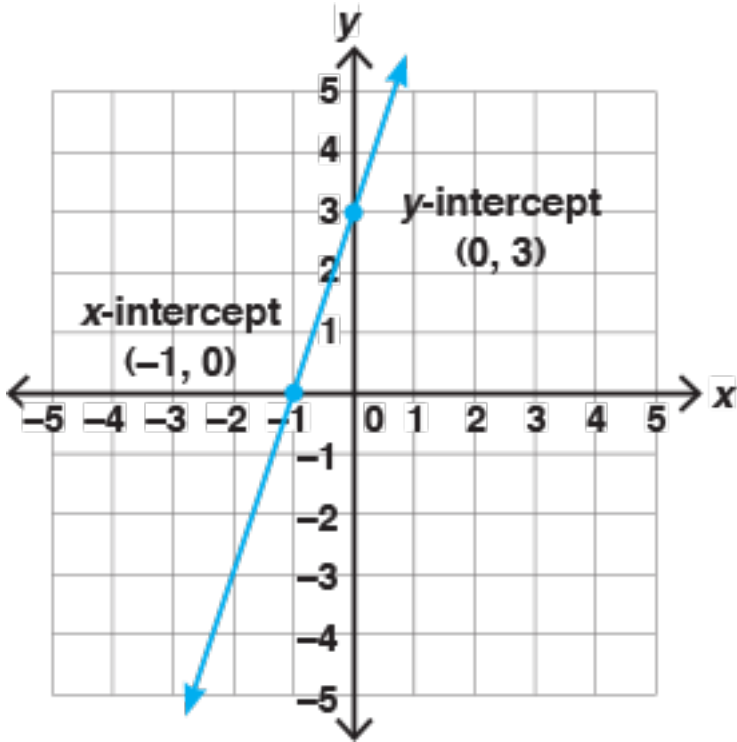
Toolbox

Using Intercepts

Using Intercepts

When a line crosses an axis, the points of intersection are called *intercepts*.

Look at the graph of $y = 3x + 3$ below.



When the line crosses the y -axis, the x -value is 0 and the y -value is 3.

The y -intercept is $(0, 3)$.

When the line crosses the x -axis, the x -value is -1 and the y -value is 0.

The x -intercept is $(-1, 0)$.

You can find the intercepts of any linear equation by substituting 0 into the equation. To find the y -intercept, substitute 0 for x . To find the x -intercept, substitute 0 for y .

Example 1

Hide

Find the x - and y -intercepts of the equation $3x - 5y = 15$.

Step 1

Find the y -intercept by substituting 0 for x .

$3(0) - 5y = 15$

$-5y = 15$

$y = -3$

$\rightarrow y = \frac{15}{-5} \rightarrow y = -3$

Step 2

Find the x -intercept by substituting 0 for y .

$3x - 5(0) = 15$

$3x = 15$

$x = 5$

$\rightarrow x = \frac{15}{3} \rightarrow x = 5$

The y -intercept is $(0, -3)$. The x -intercept is $(5, 0)$.

Intercepts make it easy to graph an equation. After you find the intercepts, plot the points on a graph and draw a line to connect them.

Example 2

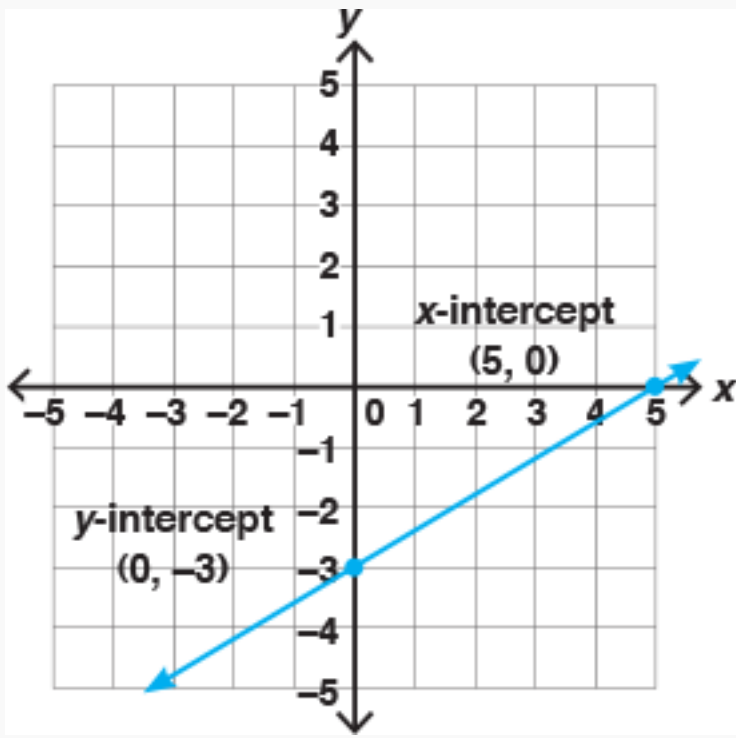
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Graph the equation $3x - 5y = 15$.

The y -intercept is $(0, -3)$.

The x -intercept is $(5, 0)$.

Plot the points and draw a line.



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Lesson 1: Graphing a Linear Equation

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